

Problem Statement (Plain English)

Learning an Interpretable Probabilistic Grammar of Hindustani Tāla — plain-English version

What is this research about, in one sentence?

We want to teach a computer to learn the "grammar" of tabla rhythm from real recordings — so it can both **create** new rhythm patterns and **explain, in a human-readable way, why each pattern is correct** — and then measure honestly what that explainability costs, if anything, compared to today's best "black box" AI models.

Some background

Tabla is a pair of hand drums central to Hindustani (North Indian classical) music. A tabla player doesn't just improvise randomly — they work within a **tāla** (a repeating rhythmic cycle) using a vocabulary of **bols**, spoken syllables like "Dha," "Ge," "Ti," "Na" that stand for specific drum strokes. Tabla players learn named patterns and operations passed down through teacher-student lineages called **gharānās** (schools) — things like a **kāyḍā** (a theme), its **palṭās** (variations on that theme), and a **tihāī** (a phrase repeated exactly three times that lands precisely back on beat one of the cycle, which requires real arithmetic to get right).

So unlike a lot of music, tabla rhythm already comes with its own **explicit vocabulary of structure** — the tradition itself names the building blocks. That makes it an unusually good subject for a computer system that is supposed to both perform and explain itself.

The problem

Today there are two separate approaches to getting a computer to produce tabla-like rhythm, and each is missing something:

1. **Hand-written rule systems** (the main historical example is a program called the Bol Processor, built starting in 1981). An expert musician sat down and wrote out the rules and their importance by hand. This produces musically sensible results, but the rules were never learned from real recordings, and were never measured against real data or compared to modern methods — there's no number you can point to that says how well these rules actually predict real performances.
2. **Modern AI models** (neural networks — the same general family of technology behind today's large language models, applied here to generating drum patterns). These are trained on real data and can produce very plausible-sounding rhythms. But they are "black boxes": ask one why it generated a particular pattern, or whether a pattern is "correct," and it cannot give a real answer — at best you can try to guess at

its reasoning after the fact using indirect techniques, which is not the same as the model actually reasoning in a way you can inspect.

No one has combined the two: a system whose rules are learned from real recordings (like the AI approach) but whose structure is still openly readable and checkable (like the hand-written approach) — and then actually measured, with real numbers, how much (if anything) you give up in quality by insisting on that transparency.

What we are building

A computer program that:

1. **Starts from real tabla structure, not guesswork.** The basic categories it uses (kāydā, dohrā, palṭā, tihāī, etc.) come from published tabla teaching material, not from the computer inventing its own categories.
2. **Learns from real recordings.** We are using a dataset of 38 tabla solo recordings, spanning 6 different gharānās, each with a syllable-by-syllable written-out score lined up in time with the audio. The program studies these recordings and learns how likely each rule is to be used, based on what actually appears in real performances — rather than a person guessing the likelihoods.
3. **Generates new rhythm patterns together with an explanation.** When the program creates a new pattern, it doesn't just hand you the pattern — it also hands you the "family tree" showing exactly which rules it applied, in order, to build that pattern. That tree is the explanation; it isn't generated separately or guessed at afterward.
4. **Checks its own arithmetic.** For tricky structures like the tihāī (three repeats that must land exactly back on beat one), the program actually verifies the timing math is correct, rather than just hoping the pattern sounds right.
5. **Gets compared fairly against modern AI, with real numbers.** We train several standard "black box" models (the same general kind of technology used in modern AI systems, scaled down and applied to this task) on the same recordings, and directly compare: which one predicts real tabla performances better? Which one's patterns look more "correct" to actual tabla players? Does having the explanation tree actually help a person do useful things — like guessing what comes next, or spotting a mistake in a pattern — better than seeing the raw pattern alone, or better than an AI model's after-the-fact guess at its own reasoning?

The central question we're trying to answer

If you insist that an AI system's reasoning be genuinely readable and checkable by a human — not just plausible-sounding — how much, if anything, do you have to give up in terms of how well it performs?

We don't know the answer in advance, and we're not assuming it will be flattering to our own approach. If the transparent system turns out to perform worse than the black-box models, that is itself a real, reportable result — it tells you what transparency costs. If it performs about the same, that's evidence you don't have to choose between an AI system you can trust and one that works well.

Four specific questions the research answers

1. **Does it actually predict real tabla music well?** Compare our system's accuracy at predicting real performances against the black-box AI models, on data none of the models have seen before.
2. **Do the rules it learns actually match what tabla teachers say?** Check whether the patterns the program considers "important" line up with the categories real tabla pedagogy already uses (kāydā, dohrā, palṭā, etc.), or whether the computer ends up inventing its own, different categories.
3. **Does the explanation actually help a person?** Run real tests with people (ideally including tabla students) to see if having the explanation tree actually improves their ability to predict, verify, or fix rhythm patterns — not just ask people if they like the explanation.
4. **What can this kind of system handle, and what can't it?** Some tabla phenomena (like the tihāī's arithmetic) need more than the basic rule system to describe correctly. This question maps out exactly which parts of tabla rhythm fit naturally into our approach, which need extra machinery bolted on, and which might not fit at all — an honest account of the limits, not just the successes.

Where the dataset side of this stands right now

We have already checked the real dataset (not just trusted its documentation) and confirmed: 38 compositions across all 6 claimed gharānās, all in the same tāla (tīntāl), and — most importantly — the written score and the timestamped recording data match up **exactly**, proving this is a genuinely trustworthy, time-aligned dataset to learn from, not just an assumption we're taking on faith. Full details are in `docs/PHASE1_DATASET_AUDIT.md` if you want the evidence.